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Academic Report: Data Federation with PostgreSQL Foreign Data Wrappers

1. Introduction

Modern enterprise systems rarely store all their data in a single monolithic database. Data is distributed across various services, formats, and physical servers. When an application needs to generate a report, analysts typically have to export data from multiple systems and join it externally, or build complex ETL pipelines. This project demonstrates an advanced database capability called **Data Federation**, enabled by PostgreSQL Foreign Data Wrappers (FDW).

2. What are Foreign Data Wrappers?

Foreign Data Wrappers are a PostgreSQL extension of the SQL/MED (Management of External Data) standard. They allow PostgreSQL to connect to external data sources—such as MongoDB, MySQL, another PostgreSQL server, or even flat files like CSVs—and query them as if they were standard relational tables residing locally.

4. Examples of FDW Wrappers

PostgreSQL includes built-in wrappers such as file_fdw and postgres_fdw, but the wider FDW ecosystem also supports many other external systems. The table below shows examples of wrappers that could be used in real-world data federation scenarios:

FDW wrapper	External source	Typical use case	Notes
file_fdw	CSV or flat files on the database server	Query exported reports or raw files without importing them	Built into PostgreSQL; read-only
postgres_fdw	Another PostgreSQL database	Join data between separate PostgreSQL databases	Built into PostgreSQL; supports reads and some writes
mysql_fdw	MySQL or MariaDB databases	Combine PostgreSQL analytics with data from MySQL applications	Requires an additional extension
oracle_fdw	Oracle databases	Access enterprise or legacy Oracle systems from PostgreSQL	Requires Oracle client libraries
tds_fdw	Microsoft SQL Server and Sybase	Federate data from SQL Server into PostgreSQL	Useful in mixed database environments
mongo_fdw	MongoDB collections	Query document-style data through PostgreSQL	Data type mapping can be more complex than relational wrappers
redis_fdw	Redis key-value data	Read cache or session-style data from PostgreSQL	Best suited for simple key-value access patterns
ogr_fdw	GIS files and spatial formats	Query shapefiles, GeoJSON, spreadsheets, and other OGR-supported sources	Often useful with PostGIS

For this project, file_fdw and postgres_fdw were selected because they are bundled with standard PostgreSQL installations and work well in a Windows development environment without requiring extra binary dependencies.

4. Data Federation Explained

Data Federation is the concept of keeping data in its original, distributed location while providing a unified, virtual view to the user.

- **Local Tables:** Data is stored physically on the disk managed by the current PostgreSQL instance.
- **Foreign Tables:** The table definition exists in PostgreSQL, but the actual data remains in the external system. When a query is executed, PostgreSQL acts as a client, fetches the necessary data from the remote source, and processes it locally.

5. Why I Chose Foreign Data Wrappers

For the technology selection, the primary constraint was the local development environment. The project had to run entirely on a personal Windows laptop with limited available storage. Extensions like TimescaleDB require downloading and manually installing additional binaries, copying DLL files into the PostgreSQL directory, and modifying system configuration files — adding several hundred megabytes of dependencies. Apache AGE is not distributed as a precompiled Windows binary and would have required compiling from source using Visual Studio C++ Build Tools, which is both storage-heavy and error-prone on Windows.

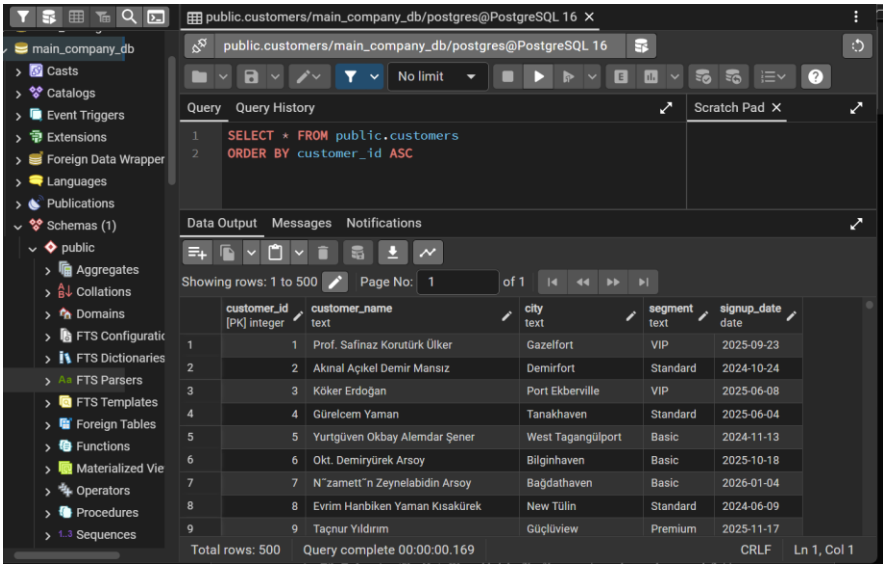
Foreign Data Wrappers, by contrast, are bundled natively with every standard PostgreSQL installation. Both file_fdw and postgres_fdw required zero additional downloads — they are activated with a single SQL command. This made FDW the most practical choice given the storage and environment constraints, while still being a genuinely advanced PostgreSQL feature not covered in the course syllabus.

Beyond practicality, FDW addresses a real and important database problem: in production systems, data rarely lives in a single database. FDW allows PostgreSQL to act as a federated query engine, joining data across a CSV file and a remote database in a single SQL statement — a capability that is directly relevant to enterprise data architectures.

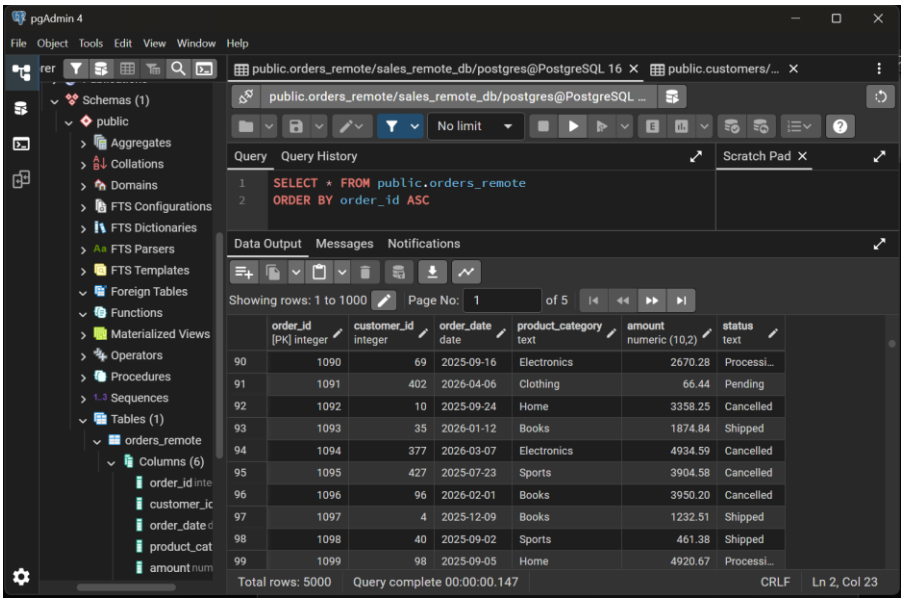
6. The Project Scenario

In our simulated enterprise scenario, we constructed an architecture utilizing three distinct data sources to represent a fragmented corporate data environment. The project uses synthetic e-commerce data consisting of 500 customers and 5,000 orders to demonstrate the database technology.

- **customers:** A local PostgreSQL table residing in the primary database (main_company_db).
- **foreign_orders_csv:** A CSV file (external_orders.csv) accessed dynamically via file_fdw.
- **orders_remote:** A table located in a completely separate database instance (sales_remote_db) accessed via postgres_fdw.



customers table



orders_remote table

7. Implementation and Execution

The project successfully established federation through the following technical steps:

1. **File Federation (file_fdw):** We enabled the file_fdw extension and created a server definition pointing to the local filesystem. We then mapped a foreign table to the CSV file, allowing PostgreSQL to parse the CSV on-the-fly during execution without importing the rows into the database storage.

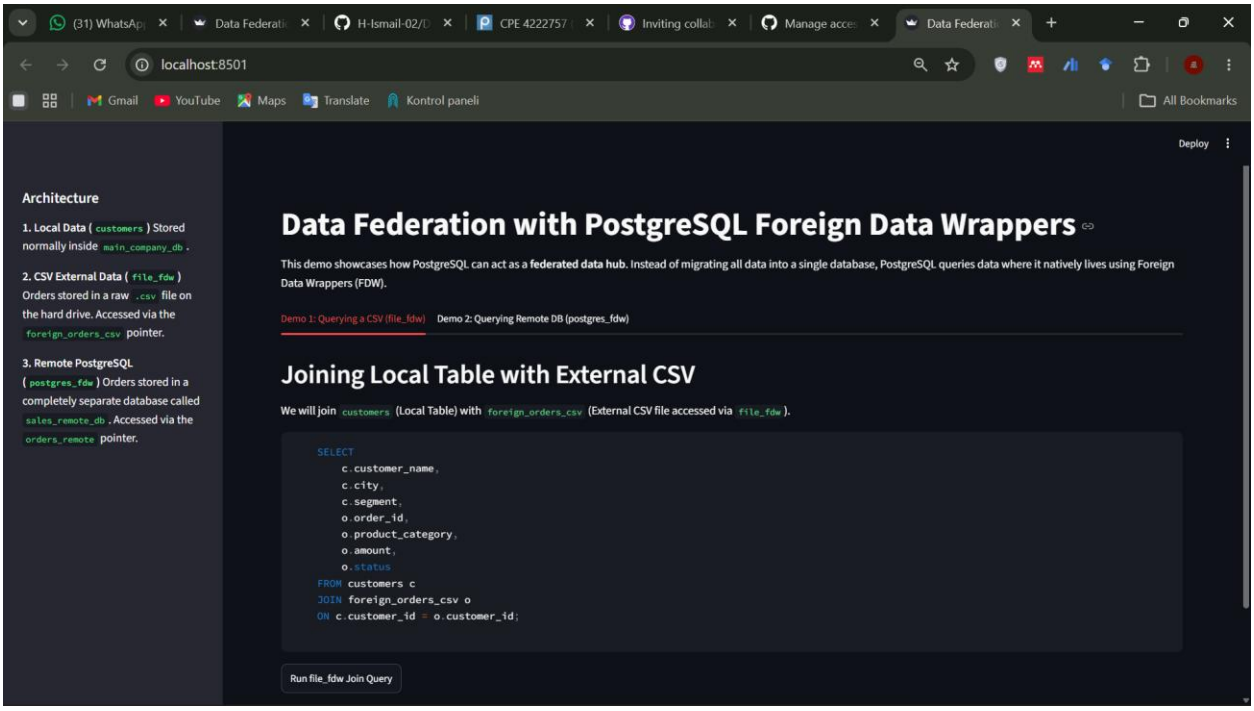
2. **Remote Database Federation (postgres_fdw):** We enabled postgres_fdw, defined the remote server (sales_remote_db), and established a user mapping to securely handle credentials. We used the IMPORT FOREIGN SCHEMA command to dynamically link the remote table definition into the local database.

3. **The Join Query:** With the architecture in place, we demonstrated the capability to perform standard SQL JOIN operations connecting the local customers table with both the CSV-backed foreign table and the remote PostgreSQL table. The query planner successfully parsed these unified queries, fetching data from disk and over the network seamlessly based on the shared customer_id.

9. Application Demonstration

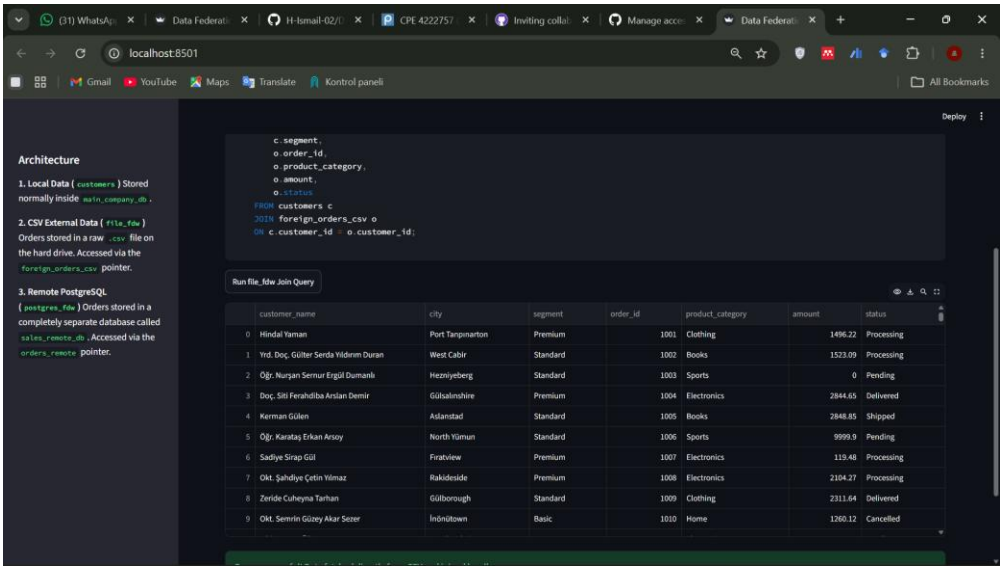
To prove the success of the data federation architecture, we built an interactive dashboard using Streamlit. This dashboard allows us to execute federated SQL JOIN queries live and observe the results.

9.1 The Federated Architecture

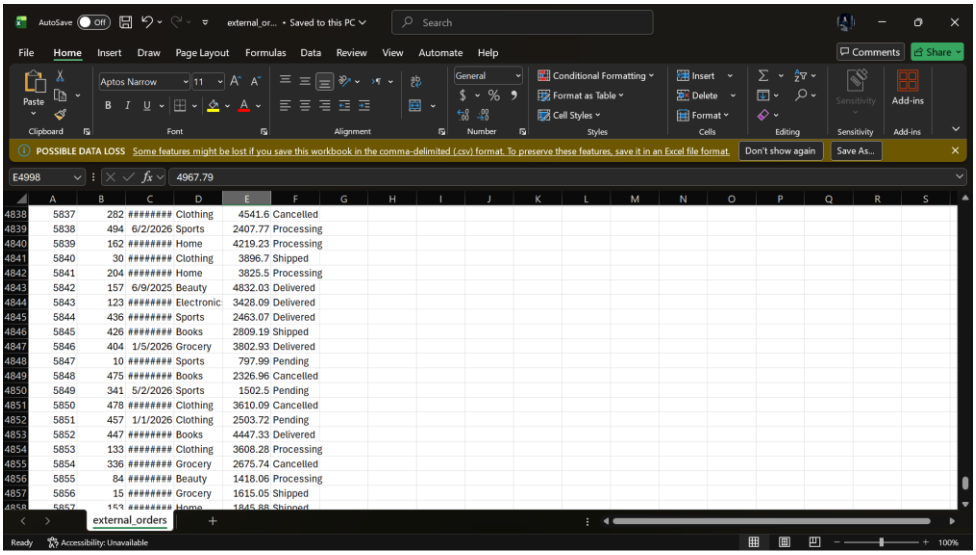


Description: The application architecture relies on three distinct storage locations. Our local PostgreSQL instance (main_company_db) stores the core customers table. However, it does not store order data. Instead, it maintains two "Foreign Table" pointers: foreign_orders_csv (pointing to a flat text file) and orders_remote (pointing to a completely separate PostgreSQL server).

9.2 Demo 1: Joining Local Data with a Text File (file_fdw)



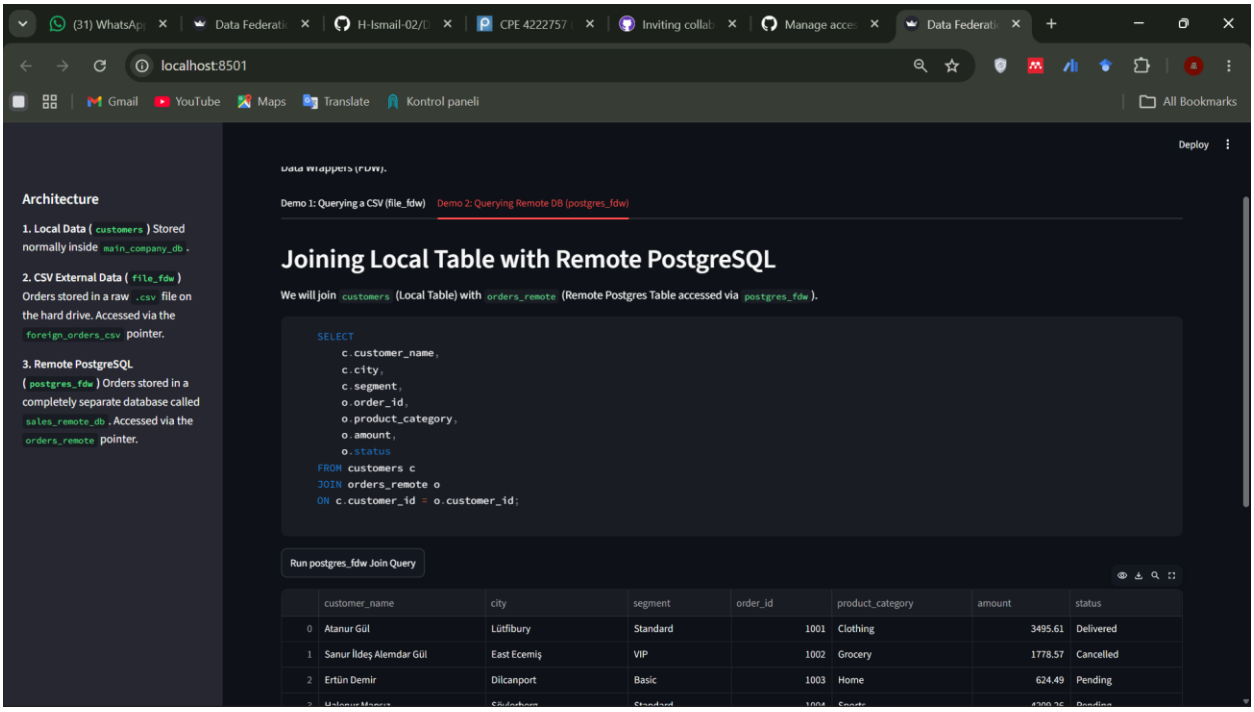
Description: In this demonstration, we executed a standard SQL JOIN command connecting our local customers table with the foreign_orders_csv pointer. Upon execution, the PostgreSQL engine instantly accessed the raw external_orders.csv text file stored on the Windows hard drive, parsed the text into structured columns in memory, and joined the records. This proves that we can query raw files dynamically without ever running an ETL script to import them.



	A	B	C	D	E	F	G
4838	5837	282	########	Clothing	4541.6	Cancelled	
4839	5838	494	6/2/2026	Sports	2407.77	Processing	
4840	5839	162	########	Home	4219.23	Processing	
4841	5840	30	########	Clothing	3896.7	Shipped	
4842	5841	204	########	Home	3825.5	Processing	
4843	5842	157	6/9/2025	Beauty	4832.03	Delivered	
4844	5843	123	########	Electronic	3428.09	Delivered	
4845	5844	436	########	Sports	2463.07	Delivered	
4846	5845	426	########	Books	2809.19	Shipped	
4847	5846	404	1/5/2026	Grocery	3802.93	Delivered	
4848	5847	10	########	Sports	797.99	Pending	
4849	5848	475	########	Books	2326.96	Cancelled	
4850	5849	341	5/2/2026	Sports	1502.5	Pending	
4851	5850	478	########	Clothing	3610.09	Cancelled	
4852	5851	457	1/1/2026	Clothing	2503.72	Pending	
4853	5852	447	########	Books	4447.33	Delivered	
4854	5853	133	########	Clothing	3608.28	Processing	
4855	5854	336	########	Grocery	2675.74	Cancelled	
4856	5855	84	########	Beauty	1418.06	Processing	
4857	5856	15	########	Grocery	1615.05	Shipped	

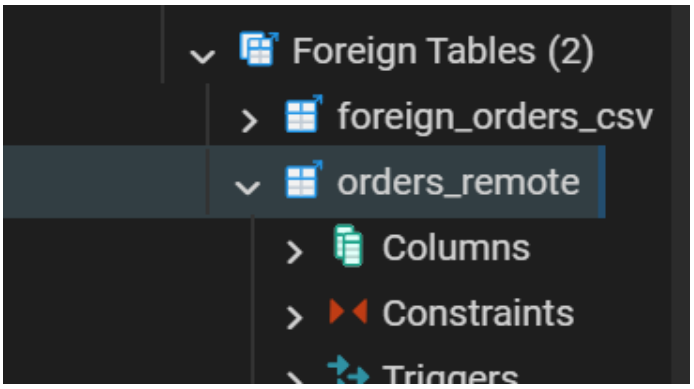
Description: By altering a value directly inside the raw .csv text file and re-running the query in our dashboard, the updated value was instantly reflected in our SQL results. This confirms the data is being read live from the disk, preventing the "stale data" issue common in traditional data pipelines.

9.3 Demo 2: Joining Local Data with a Remote Database (postgres_fdw)



Description: In our second demonstration, we executed a JOIN between our local customers table and orders_remote. In the background, our primary database acted as a client, reached out over the network to a completely independent database instance (sales_remote_db), fetched the requested order data, and merged it locally. This illustrates how PostgreSQL can act as a unified hub for a fragmented corporate ecosystem.

9.4 Verification in pgAdmin



FOREIGN TABLES in main_company_db

Description: To verify the physical separation of data, we inspected the database schemas using pgAdmin. As shown above, the orders_remote and foreign_orders_csv entities are listed strictly under the **Foreign Tables** directory and have a specialized icon. They do not exist as physical tables inside the main database, definitively proving that data federation was successfully achieved.

10.Technology Stack

To construct this federated architecture and the interactive dashboard, the following technology stack was utilized:

- **PostgreSQL 16:** The core open-source Relational Database Management System (RDBMS). In this project, it was utilized not just as a storage mechanism, but as the central federated query engine.
- **Foreign Data Wrappers (SQL/MED):** Native PostgreSQL extensions used to establish the federated links.
    - **file_fdw**: Utilized to parse and query raw CSV text files from the Windows local file system in real-time.
    - **postgres_fdw**: Utilized to establish a network connection to an independent, remote PostgreSQL server to query data dynamically.
- **Python 3 & psycopg2:** The backend programming language and database adapter used to programmatically automate the DDL (Data Definition Language) execution and configure the FDW environments.
- **Pandas & Faker:** Python libraries used to algorithmically generate 5,500 rows of highly realistic synthetic e-commerce data. This ensured the architecture could be tested without relying on static public datasets.
- **Streamlit:** A rapid web application framework used to construct the interactive frontend dashboard. This was crucial for proving that end-user applications can execute federated SQL JOIN queries dynamically as if they were querying a standard local database.

11. Advantages, Limitations, and Security

Advantages

- **No Data Duplication:** Reduces storage costs and eliminates the risk of outdated ETL pipelines.
- **Real-Time Access:** Queries against foreign tables fetch the most up-to-date data directly from the source.
- **Simplicity:** Developers can write standard SQL JOINs instead of complex application-level API routing or integration logic.

Limitations

- **Performance:** Since data must be transferred over the network (or read sequentially from disk for CSVs) during the query execution, it is generally slower than querying heavily indexed local tables.
- **Write Constraints:** While postgres_fdw supports INSERT, UPDATE, and DELETE operations back to the remote server, extensions like file_fdw act strictly as read-only interfaces.

Security Considerations

When using postgres_fdw, user mappings must be securely configured to prevent unauthorized access. The credentials for the remote database are stored inside the primary database's system catalogs and must be restricted using careful role-based access control (RBAC).

12. Conclusion

This project successfully demonstrates that PostgreSQL is not just a relational database, but a powerful data federation hub. By utilizing Foreign Data Wrappers, organizations can abstract away the complexity of distributed systems and treat disparate data sources as a single, unified relational schema.